Shourya Pratap Singh

CS Junior at VIT Chennai

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Education

BTech in Computer Science and Engineering | *VIT Chennai* | *October 2022 - July 2026* | *CGPA:* 8.70/10
Class 12 | *CBSE* | *The Orbis School, Pune* | *June 2020 - July 2022* | *Score:* 94.80%
Class 10 | *ICSE* | *The Bishop's School, Pune* | *June 2007 - June 2020* | *Score:* 96.80%

Experience

1) Data Science Intern, Baker Hughes – CarbonEdge 01/2026–06/2026

- Worked on injectivity optimisation for Carbon Capture, Utilization and Storage (CCUS) workflows using pressure-temperature (PT) gauge data and time-series forecasting techniques.
- Built time-series forecasting pipelines for injectivity optimisation in CarbonEdge using Temporal Transformers and regression-based statistical modelling on PT gauge data. Achieved reduction in forecast noise by 50%.
- Collaborated cross-functionally with reservoir, petroleum and geology experts to support data-driven operational analysis. Used: Python, NumPy, Pandas, PyTorch.

2) Machine Learning (ML) Intern, Center for Development of Advanced Computing 05/2025–07/2025

- Interned at CDAC Bioinformatics Group at Pune under Dr. Sunitha Manjari Kasibhatla & Dr. Archana Achalere.
- Engineered an ML model to perform unsupervised segmentations from given Breast Cancer whole slide images, highlighting region of interest with dice scores exceeding 0.9. Implemented OpenCV techniques, Vision Transformers, Graph Neural Networks, Multiple Instance Learning and previous benchmark papers such as CUTS and UnSegArmaNet. Used: Python, Pytorch.

3) Machine Learning (ML) Research Intern, National Taiwan University 05/2024–07/2024

- Interned at the Computational Intelligence in Biomedical Imaging lab at NTU under Prof. Cheng Ying Chou.
- Engineered an ML model to detect colon cancer from CT scans, using UNets(Convolutional Neural Networks) and EffTrans (Transformer), achieving 91.5% diagnostic accuracy in detecting colon cancer, surpassing previous lab benchmarks by more than 5%. Worked with Python, Pytorch, CuDNN and CUDA.

Publications

1) A review on rhodamine probes for metal ion recognition with a future on AI and ML
Published in Elsevier - Coordination Chemistry Reviews (Journal Impact Factor-24.83) Paper Link

- Worked in a multidisciplinary team under the guidance of Dr. Pritam Ghosh on developing an ML tool for image analysis in chemistry. Developed the model using chemosensing datasets to analyze analytes in unknown specimens. Developed it using TensorFlow and Python-based OpenCV, CNN, and SVM.
- The publication has been cited 35 times since publication. (As of April 2025).

Projects

sps-crypto - A Python package covering all aspects of foundational cryptography PyPI Link
Tech: Python, PyPI

- Created a Python package which includes programs to perform cryptographic actions, including ciphers, encryptions, signatures, key exchanges, and other math and networking aspects related to digital cryptography. Built as a course project. Wrote unit tests, did unit, integration testing, and handled version control with open source contributions toward the package.

AMKR - Assistive Eyewear for Visually Impaired *Product*
Tech: Raspberry Pi 0, PiCam V3, Text-to-Speech (TTS) API, GPT-4 Vision API, GoogleFlow Lite (TFLite), Microsoft Azure

- Designed eyewear to help the blind read text, recognize faces, and perceive the scene. Utilized GenAI tools such as Meta NougatOCR and GPT-4 Vision API for to enhance optical character recognition; OpenAI Text-to-Speech (TTS) for auditory feedback; Google TensorFlow Lite(TFLite) for face recognition.

Positions of Responsibility

1) Research Lead, Dr. Ibrahim Research Group, Data Science Club VIT Chennai | 2024-2025
2) Student Chair, Campus Development Committee, VIT Chennai | 2023-2024
3) Head of Finance, DevsHouse '24, VIT Chennai |2023-2024

Skills

- Pytorch, Tensorflow, Linux, CUDA, CUDnn, Python, C++; Transformers, CNNs, UNets, ANNs, and RNNs,
- Python and C++ for ML and Computer Vision; Flutter, React Native, and Firebase for app development; Node for backend; HTML5, CSS, JS, React for Web Development.
- Heroku and Google Cloud Platform to host projects and products developed. Worked with Kubernetes for distributed systems. Experienced with Cisco Packet Tracer and networks programming (TCP/IP).
- Certified in networks programming by Cisco and Cloud computing by Google.

Achievements

- Google Build with AI, Bangalore Project Demonstration - 2024
- 2x VNEST - Intra VIT Hackathon winner - 2023
- IEEE YESIST12 Bangalore section winner - 2023
- Top 15 - CyberX Hackathon, Greater Chennai Police - 2023
- Times NIE, Star Correspondent, Pune 2018-2024.